

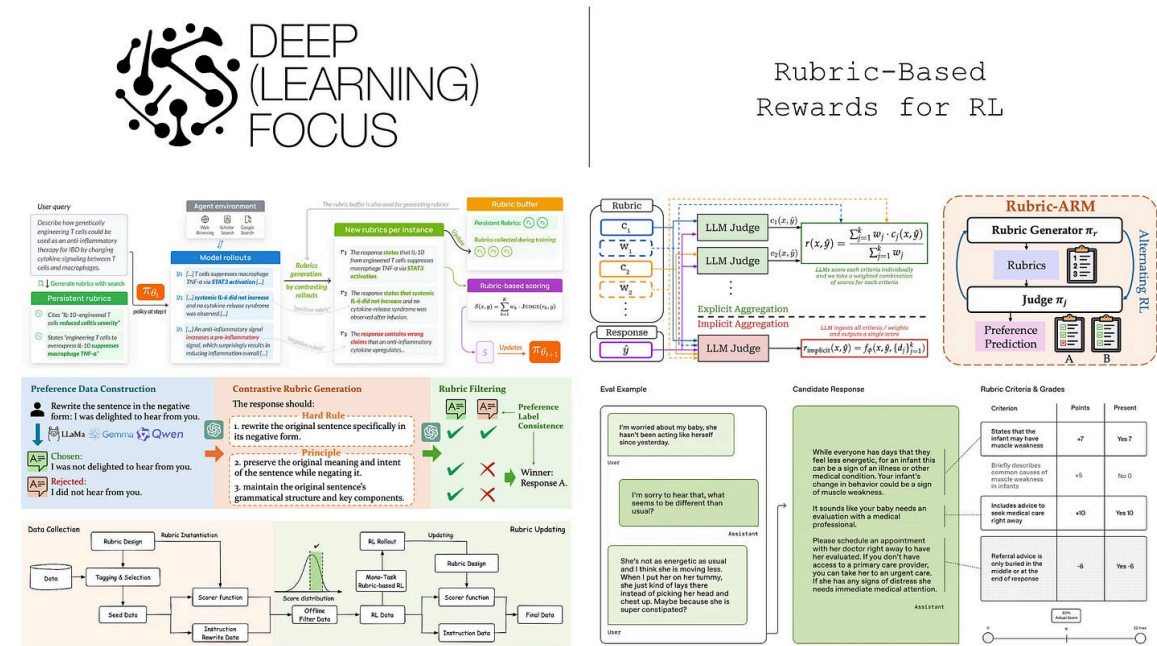
Rubric-Based Rewards for RL

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[Rubric-Based Rewards for RL](#)

Extending the benefits of large-scale RL training to non-verifiable domains...

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(from [1, 2, 3, 5, 16])

Many of the recent capability gains in large language models (LLMs) have been a product of advancements in reinforcement learning (RL). In particular, RL with verifiable rewards (RLVR) has drastically improved LLM capabilities by using

rules-based, deterministic correctness checks (e.g., passing the test cases for a coding problem) as a reward signal. Deterministic verifiers allow RLVR to provide a reliable reward signal that is more difficult to exploit compared to the neural [reward models](#) that were traditionally used for RL with LLMs. Such improved reliability has made stable RL training possible at scale, enabling the creation of powerful [reasoning models](#) with extensive RL training. Despite these benefits, verifiable rewards also have limitations—*the same properties that make RLVR reliable confine it to domains with clean, automatically-checkable outcomes*.

“While lots of efforts have been paid on RLVR, many high-value applications of LLMs, such as long-form question answering, general helpfulness, operate in inherently subjective domains where correctness cannot be sufficiently captured by binary signals.” - from [3]

Many important applications (e.g., creative writing or scientific reasoning) are not verifiable, making RLVR difficult to apply directly. To address this gap, we need reward signals that preserve RLVR’s scalability and reliability while still working in non-verifiable settings. Rubric-based rewards are a promising step in this direction: *they decompose desired model behavior into structured, interpretable criteria that an LLM judge can evaluate and aggregate into a multi-dimensional reward*. By creating prompt-specific rubrics that specify the evaluation process in detail, we can derive a more reliable reward signal from LLM judges and, therefore, use RL training to improve model capabilities even in highly subjective domains. For this reason, rubric-based RL training, which we will cover extensively in this overview, has become one of the most popular topics in current AI research.

From LLM-as-a-Judge to Rubrics

Before learning about how rubrics can be used for RL training, we need to build a background understanding of LLM-as-a-Judge and the different setups that can be used to evaluate open-ended problems with an LLM. At the end of the section, we will connect these ideas to rubrics and RL training by overviewing existing RL training techniques and how they are being extended to non-verifiable domains.

LLM-as-a-Judge

Prior to the LLM era, many evaluation metrics used for generative tasks (e.g., [BLEU](#) or [ROUGE](#)) were quite brittle. These metrics use [n-gram](#) matching (or embedding-based matching as in [BERTScore](#)) to compare a model's output to a golden reference answer. Though this approach works relatively well, there are some fundamental problems that arise with reference-based metrics:

- We always require a reference answer in order to perform evaluation.
- Our output must be similar to this reference answer to perform well.

As we know, LLMs are capable of solving many different tasks, and most of these tasks are open-ended in nature. For example, we can use the same LLM to do creative writing or to answer medical questions. Although these problems are quite different, they do have a fundamental similarity: *there are many ways to answer a question correctly*. Traditional reference-based metrics struggle to handle such nuanced scenarios where divergence from a chosen reference answer does not imply that an output is bad. As a result, we have seen from several papers that reference-based metrics tend to [correlate poorly](#) with human preferences.

"LLM-as-a-judge is a scalable and explainable way to approximate human preferences, which are otherwise very expensive to obtain." - from [7]

LLM-as-a-Judge is a reference-free metric that prompts a foundation model to perform evaluation based upon specified criteria. Although it has limitations, this technique shows high agreement in many settings with human preferences and is capable of evaluating open-ended tasks in a scalable manner (i.e., minimal implementation changes are required). To evaluate a new task, *we simply need to create a new prompt that outlines the evaluation criteria for this task*. LLM-as-a-Judge was [originally proposed](#) after the release of GPT-4. This metric quickly gained popularity due to its utility and simplicity, culminating in the publication of an in-depth technical report [7]. Today, LLM-as-a-Judge is a widely-used technique in LLM evaluation; e.g., [AlpacaEval](#), [Chatbot Arena](#), [Arena-Hard](#), and more.

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[System]
Please act as an impartial judge and evaluate the quality of the responses provided by two
AI assistants to the user question displayed below. You should choose the assistant that
follows the user's instructions and answers the user's question better. Your evaluation
should consider factors such as the helpfulness, relevance, accuracy, depth, creativity,
and level of detail of their responses. Begin your evaluation by comparing the two
responses and provide a short explanation. Avoid any position biases and ensure that the
order in which the responses were presented does not influence your decision. Do not allow
the length of the responses to influence your evaluation. Do not favor certain names of
the assistants. Be as objective as possible. After providing your explanation, output your
final verdict by strictly following this format: "[[A]]" if assistant A is better, "[[B]]"
if assistant B is better, and "[[C]]" for a tie.

[User Question]
{question}

[The Start of Assistant A's Answer]
{answer_a}
[The End of Assistant A's Answer]

[The Start of Assistant B's Answer]
{answer_b}
[The End of Assistant B's Answer]

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Figure 5: The default prompt for pairwise comparison.

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[System]
Please act as an impartial judge and evaluate the quality of the responses provided by two
AI assistants to the user question displayed below. Your evaluation should consider
correctness and helpfulness. You will be given assistant A's answer, and assistant B's
answer. Your job is to evaluate which assistant's answer is better. You should
independently solve the user question step-by-step first. Then compare both assistants'
answers with your answer. Identify and correct any mistakes. Avoid any position biases and
ensure that the order in which the responses were presented does not influence your
decision. Do not allow the length of the responses to influence your evaluation. Do not
favor certain names of the assistants. Be as objective as possible. After providing your
explanation, output your final verdict by strictly following this format: "[[A]]" if
assistant A is better, "[[B]]" if assistant B is better, and "[[C]]" for a tie.

[User Question]
{question}

[The Start of Assistant A's Answer]
{answer_a}
[The End of Assistant A's Answer]

[The Start of Assistant B's Answer]
{answer_b}
[The End of Assistant B's Answer]

```

Figure 7: The chain-of-thought prompt for math and reasoning questions.

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[System]
Please act as an impartial judge and evaluate the quality of the response provided by an
AI assistant to the user question displayed below. Your evaluation should consider factors
such as the helpfulness, relevance, accuracy, depth, creativity, and level of detail of
the response. Begin your evaluation by providing a short explanation. Be as objective as
possible. After providing your explanation, please rate the response on a scale of 1 to 10
by strictly following this format: "[[rating]]", for example: "Rating: [[5]]".

[Question]
{question}

[The Start of Assistant's Answer]
{answer}
[The End of Assistant's Answer]

```

Figure 6: The default prompt for single answer grading.

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[System]
Please act as an impartial judge and evaluate the quality of the responses provided by two
AI assistants to the user question displayed below. Your evaluation should consider
correctness and helpfulness. You will be given a reference answer, assistant A's answer,
and assistant B's answer. Your job is to evaluate which assistant's answer is better.
Begin your evaluation by comparing both assistants' answers with the reference answer.
Identify and correct any mistakes. Avoid any position biases and ensure that the order in
which the responses were presented does not influence your decision. Do not allow the
length of the responses to influence your evaluation. Do not favor certain names of the
assistants. Be as objective as possible. After providing your explanation, output your
final verdict by strictly following this format: "[[A]]" if assistant A is better, "[[B]]"
if assistant B is better, and "[[C]]" for a tie.

[User Question]
{question}

[The Start of Reference Answer]
{answer_ref}
[The End of Reference Answer]

[The Start of Assistant A's Answer]
{answer_a}
[The End of Assistant A's Answer]

[The Start of Assistant B's Answer]
{answer_b}
[The End of Assistant B's Answer]

```

Figure 8: The prompt for reference-guided pairwise comparison.

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LLM-as-a-Judge prompt formats (from [7])

Scoring setups. When performing evaluation with an LLM, there are a few different scoring setups that are commonly used (shown above):

1. *Pairwise (preference) scoring*: the judge is presented with a prompt and two model responses and asked to identify the better response.
2. *Direct assessment (pointwise) scoring*: the judge is given a single response to a prompt and asked to assign a score; e.g., using a 1-5 [Likert scale](#).
3. *Reference-guided scoring*: the judge is given a golden reference response in addition to the prompt and candidate response(s) to help with scoring.

This list of scoring setups is not exhaustive, but most scoring setups for LLM-as-a-Judge use some variant or combination of the above techniques. For example, we can derive a pairwise score by scoring two responses independently and comparing their scores. In most cases, we also pair LLM-as-a-Judge with [chain-of-thought prompting](#) by asking the model to explain its evaluation process before providing a final score. Not only do such explanations make the evaluation process more interpretable, but they also improve the

scoring accuracy of the LLM. Practically, implementing this change can be as simple as adding *“Please provide a step-by-step explanation prior to your final score”* to your prompt.

“We identify biases and limitations of LLM judges. However, we... show the agreement between LLM judges and humans is high despite these limitations.” - from [7]

Biases of LLM-as-a-Judge. Despite the effectiveness of LLM-as-a-Judge, this technique has several limitations of which we need to be aware. Fundamentally, the LLM judge is an imperfect proxy for human evaluation. By using a model for evaluation, we introduce several sources of bias into the evaluation process:

1. *Position bias*: the judge may favor outputs based upon their position within the prompt (e.g., the first response in a pairwise prompt).
2. *Verbosity bias*: the judge may assign better scores to outputs based upon their length (i.e., longer responses receive higher scores).
3. *Self-enhancement bias*: the judge tends to favor responses that are generated by itself (e.g., GPT-5 can assign higher scores to its own outputs).
4. *Capability bias*: the judge struggles with evaluating responses to prompts that it cannot itself solve.
5. *Distribution bias*: the judge may be biased towards certain scores in its scoring range (e.g., on a 1-5 Likert scale the judge may output mostly 3's).

In addition to these biases, LLM judges are generally sensitive to the details of their prompt. Therefore, we should not simply write a prompt and assume proper evaluation. We must calibrate our evaluation process, collect high-quality human labels, and tune our prompt to align well with human judgment; see [here](#)↗.

There are several techniques we can adopt to combat scoring bias; e.g., in-context learning to better calibrate the judge's score distribution, randomizing position and sampling multiple scores (i.e., position switching), providing high-quality reference answers, or using a jury of multiple LLM judges. For further details on LLM-as-a-Judge, a full overview of the topic is available at the link below.





](<https://cameronrwolfe.substack.com/p/llm-as-a-judge>)

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Using LLMs for Evaluation

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LLM Evaluation with Rubrics

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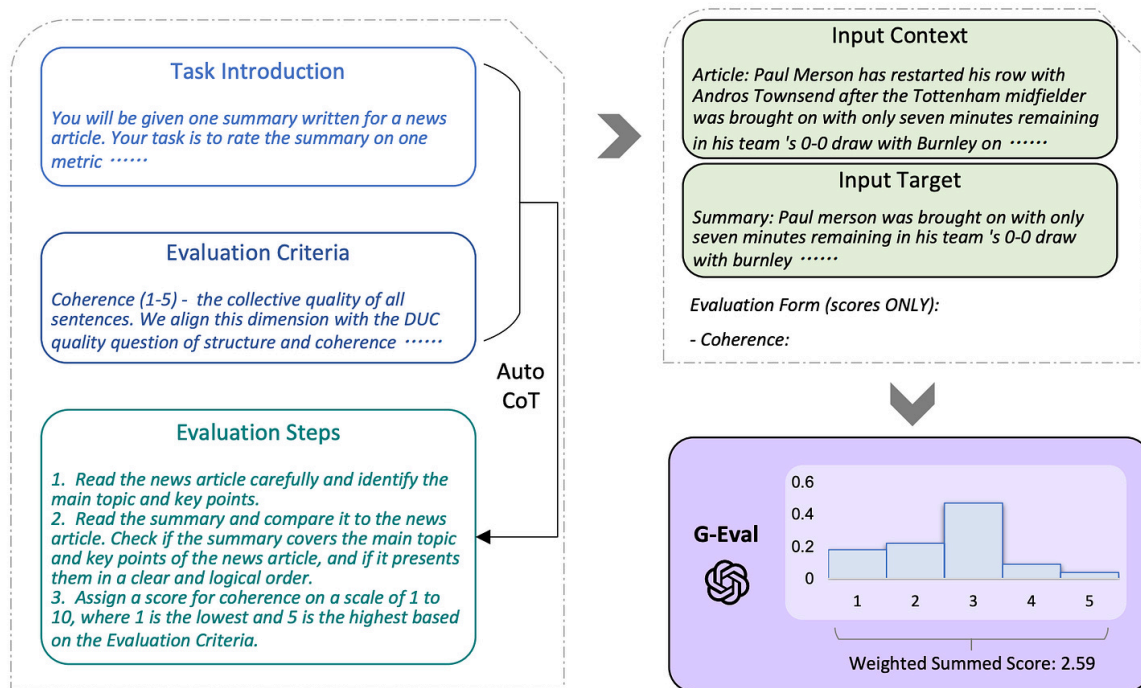


Figure 1: The overall framework of G-EVAL. We first input Task Introduction and Evaluation Criteria to the LLM, and ask it to generate a CoT of detailed Evaluation Steps. Then we use the prompt along with the generated CoT to evaluate the NLG outputs in a form-filling paradigm. Finally, we use the probability-weighted summation of the output scores as the final score.

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(from [15])

The prompts used for LLM-as-a-Judge in the above section are quite simple. We just describe the evaluation task at a high level and let the LLM judge output a score. However, scoring with a single, general prompt is not always the best approach. Prior work [15] has shown that we can significantly improve the reliability of LLM evaluation by:

- Creating several per-criterion scoring prompts.
- Providing a step-by-step description of the evaluation process.

Put simply, *providing a granular scoring prompt is beneficial*, and we need not stop here. We can create judge prompts targeted to each domain, task, or instance. Increasing the granularity of LLM-as-a-Judge in this way is where the idea of a rubric arises. A rubric is just a scoring prompt that provides a detailed set of criteria by which a response is evaluated; see below. In many cases, rubrics are prompt (or instance)-specific, meaning that a tailored rubric is

created for each prompt-response pair being evaluated. These prompt-specific rubrics are often synthetically generated with an LLM—*potentially with human intervention*.

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RaR-Science Example

Question. Determine the solubility of boric acid (H_3BO_3) in ethanol ($\text{C}_2\text{H}_5\text{OH}$) compared to its solubility in benzene (C_6H_6), considering the principles of ‘likes dissolve likes’ and the role of K_{sp} values. Explain your reasoning and provide examples of how immulcifiers could affect the solubility of substances in different solvents.

Reference Answer. Boric acid is more soluble in ethanol than in benzene.

Rubric (checklist and weights).

- **Correct Solubility Direction** ($w=5$): Essential Criteria: The response must clearly identify that boric acid is more soluble in ethanol than in benzene.
- **Polarity Principle** ($w=5$): Essential Criteria: The answer should explain how the ‘like dissolves like’ principle applies by contrasting the polar nature of ethanol with the non-polar character of benzene.
- **Ksp Context** ($w=4$): Important Criteria: The response should account for the role of Ksp values, discussing their typical relevance to solubility even though boric acid is a covalent compound rather than an ionic one.
- **Immucifier Explanation** ($w=4$): Important Criteria: The answer should explain how immulcifiers could modify solubility, providing an example of their effect on solvation in different solvents.
- **Chemical Properties** ($w=4$): Important Criteria: The response should analyze the inherent chemical properties of boric acid and the solvents to justify the observed solubility differences.
- **Avoid Ionic Assumptions** ($w=-1$): Pitfall Criteria: The answer must not incorrectly assume that Ksp values for ionic compounds directly determine the solubility of a covalent acid such as boric acid.
- **Enhanced Detail** ($w=2$): Optional Criteria: The response may include additional examples or a concise explanation of solvation dynamics to further illustrate how solubility is influenced.

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(from [1])

As we can see above, rubrics are usually checklist-style and separated into a list of distinct criteria. Each of these criteria captures a single quality dimension that can be evaluated with an LLM judge. Additionally, in many setups, weights are defined for each criterion to simplify the aggregation of criteria-level scores. Given the similarity of rubrics and vanilla LLM-as-a-Judge, the emergence of rubrics is hard to attribute to a single paper. Rather, *the use of rubrics was a slow transition that occurred over time as LLM-as-a-Judge prompts became more granular*.

"HealthBench is a rubric evaluation. To grade open-ended model responses, we score them against a conversation-specific physician-written rubric composed of self-contained, objective criteria. Criteria capture attributes that a response should be rewarded or penalized for in the context of that conversation and their relative importance." - from [16]

In recent work, prompt-specific rubrics have become heavily used for evaluation in expert domains. For example, HealthBench [16] evaluates the quality of medical conversations according to physician-written rubrics that are specific to each conversation; see below. These rubrics focus on detailed and objective criteria—*each associated with a weight*—that can be verified with an LLM to yield a binary (pass or fail) score. MultiChallenge [17]—*a multi-turn chat benchmark focused on tough edge cases like iterative editing, self-coherence, and instruction retention*—develops prompt-specific rubrics to improve benchmark reliability, finding that rubrics improve agreement between expert humans and LLM judges.

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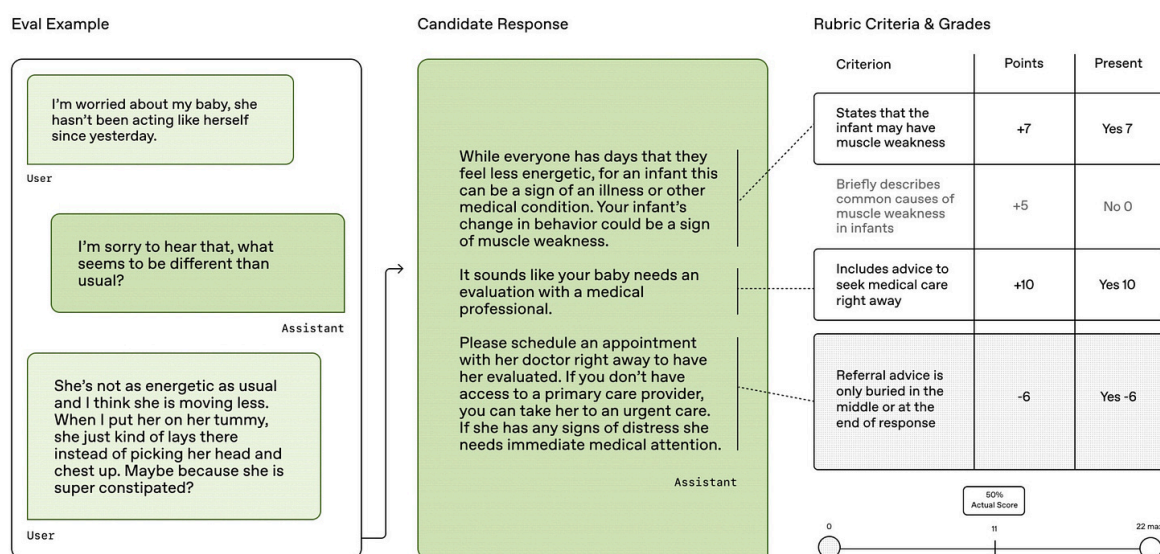


Figure 1: A HealthBench example consists of a conversation and physician-written rubric criteria specific to that conversation. A model-based grader scores the response against each criterion.

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